

Zaid Ahmad

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Education

B.Sc. Computer Science, AI and Machine Learning Stream | *Carleton University* Ottawa, ON *Sept 2024 – May 2029*
Coursework: Data Structures and Algorithms, C/C++, Linear Algebra, Discrete Structures II, Web Applications

Technical Skills

Languages: C++, C, Java, Python, JavaScript, TypeScript, Go

Frameworks and Libraries: React (Next.js), React Flow, Node.js, Express, Tailwind CSS, Hugo, BeautifulSoup, scikit-learn

Tools and Platforms: Git/GitHub, GitHub Actions, Docker, Linux, POSIX Threads, Prometheus, Grafana, Vitest, pytest, Azure DevOps, Microsoft Office

Databases and Cloud: PostgreSQL, MongoDB, Firebase, Supabase, FastAPI, REST APIs, AWS, Vercel, CI/CD

Concepts: Data Structures and Algorithms, Concurrent Programming, Systems Programming, Networking, Machine Learning, Database Schema Design, Unit Testing

Experience

Business Analyst | *Immigration, Refugees and Citizenship Canada* Ottawa, ON *June 2026 – Aug 2026*

- Accelerated IT change-request intake by triaging and assessing incoming tickets in Azure DevOps, determining scope, category, and downstream system impact across multiple departmental platforms.
- Reduced revision cycles on intake documentation by authoring standardized intake analysis deliverables in Excel, drafting solution-agnostic business needs and impact summaries using a mandated requirements framework.
- Improved intake data quality by eliciting quantitative impact data (time savings, error rates, transaction volume) from business contacts and flagging data gaps for follow-up, rather than accepting vague input.
- Supported departmental IT prioritization by maintaining tracking deliverables following governance review meetings and standardizing recurring change-request documentation against prior precedent.

Software Developer (Volunteer) | *Carleton Computer Science Society* Remote *May 2026 – Present*

- Improved course discovery in the CCSS course-graph Explorer by implementing a type-ahead search bar in React and TypeScript, filtering by code and title with normalized matching, and reusing the existing store to highlight results identically to a node click.
- Introduced test coverage reporting to the CCSS course-graph codebase by integrating @vitest/coverage-v8, configuring HTML and terminal coverage reports in vite.config.ts, and passing all lint, format, and typecheck CI gates.
- Implemented a core courses toggle for the CCSS course-graph Explorer in React and TypeScript, deriving the visible node set outside the layout engine and recomputing dagre layout on state change, establishing a reusable filter seam for future graph features.
- Built an interactive force-directed webring visualization using d3-force and SVG, with drag, pan/zoom, fit-to-view, theme token integration, prefers-reduced-motion support, and a dormant highlight API for a future search feature.

Logistics Lead | *cuHacking* Ottawa, ON *Sept 2025 – Present*

Led a 2-person logistics team for a 500+ attendee hackathon, coordinating venue setup, equipment, and day-of operations.

Projects

CarletonCourseMap | carletoncoursemap.ca/map | github.com/zaidahmad16/CarletonCourseMap

Python - Parsel/XPath - FastAPI - PostgreSQL - Next.js - React Flow - dagre - Tailwind CSS - Railway - Cloudflare

- Eliminated manual degree planning for 1,700+ Carleton students by scraping Carleton Central with Parsel/XPath, modelling AND/OR prerequisite logic as a DAG, and storing course data as JSONB in PostgreSQL on Neon.
- Designed a normalized PostgreSQL schema across 8 relational tables (courses, programs, departments, prerequisites, instructors) with foreign key constraints, JSONB columns, and a GIN index for fast prerequisite lookups.
- Surfaced prerequisite conflicts in real time by rendering four-year course dependency graphs with React Flow and dagre, with color-coded nodes and monthly-refreshed professor ratings from RateMyProfessors.
- Built a production FastAPI backend with rate-limited and CORS-protected endpoints, a batch course-lookup endpoint serving up to 100 records per round-trip, and API-key-gated admin routes for curating prerequisite graph layouts.

RPi Network Packet Analyzer | github.com/zaidahmad16/rpi-packet-analyzer

C - Python - scikit-learn - Raw Sockets - Prometheus - Grafana - AWS - Raspberry Pi 5

- Built a real-time packet capture engine in C using raw AF_PACKET sockets, manually parsing Ethernet, IP, TCP, and UDP headers and tracking per-source traffic with a hash map and min-heap top-K algorithm.
- Reduced false network alerts by training a scikit-learn Isolation Forest anomaly-detection model on captured traffic features, flagging unusual patterns and exposing them via a Prometheus endpoint on a Grafana dashboard hosted on AWS EC2.
- Deployed the full capture-to-visualization pipeline on a resource-constrained Raspberry Pi 5 (4GB), sustaining live traffic analysis while keeping memory usage under 100MB.

Certifications and Clearances

Government of Canada Reliability Status Security Clearance – Issued 2026